

Engineering Experience

1. Describe a skill or knowledge you acquired recently that has been impactful for you. Why did you make this investment? What has the outcome been for you and your team?

Recently, I have learned **PyTorch** and still deepening my understanding to improve my skills in building and experimenting with machine learning models. Recently I have realized that AI/ML is popular and the very first language I learnt in my life is my python so I thought to give it a try and it was worth it also I am having hands-on knowledge of tensor operations and gradient-based learning systems. PyTorch stood out due to its flexibility and Pythonic design, which made it easier to prototype and debug deep learning models. NumPy, on the other hand, helped me solidify my understanding of vectorized operations, broadcasting, and matrix algebra, these concepts helped me build foundation in any AI or data science work.

2. What new skill would you like to learn? Why do you think this is important or timely or interesting? Why do you think you will be good at it?

A skill I recently focused on is deepening my understanding of **NumPy** and its role in efficient numerical computing with Python. While I had used it superficially before, I made a conscious effort to go deeper and exploring array broadcasting, slicing techniques, memory optimization, and vectorized operations. I made this investment because I realized how foundational NumPy is, not just for machine learning frameworks like PyTorch, but also for writing fast, clean, and efficient code in any data-heavy application. I want to deepen my understanding of systems-level programming and Linux kernel internals. This is particularly important in environments like Canonical's, where performance and resource control often intersect with user-space tools.

3. What kinds of software projects have you worked on before? Which operating systems, development environments, languages, databases?

I've worked on a range of projects including an AI-powered multi-agent orchestration tool, a frontend-based algorithm visualizer, and infrastructure security configurations during my internships. All of my development so far has been on **Windows**, which has been my primary operating system throughout college and internships. I've used **Visual Studio Code** as my main editor, along with **Git**, **Postman**, and **Docker Desktop** for version control, API testing, and containerization workflows. I'm most experienced in **Python**, **JavaScript**, **C++**, and **TypeScript**, and I've worked with frameworks like **React.js** on the frontend and **PyTorch** for AI and numerical tasks. For databases, I've primarily used **SQL** in academic projects and lightweight formats like **JSON** in frontend and API-based applications. Most of the tools I've used are cross-platform, which has allowed me to collaborate with others regardless of their environment and adapt to project needs smoothly.

4. Why Python, and what's your history with this language?

Python was one of the first languages I learned in school, and it immediately stood out for its simplicity and readability. Over the years, I've used it in several contexts from scripting and automation to cloud configurations and AI SDK integrations. I appreciate how it allows me to write clean, maintainable code quickly, and how rich its ecosystem is in terms of libraries, especially in DevOps, cloud, data science, and machine learning. In my opinion Python is the best language among all and in future **Python will be helpful to build more advanced AI and machine learning (ML) systems.**

5. Would you describe yourself as a high quality coder? Why?

Yes, I would. I strive for code that's not only functional but also clean, modular, and well-documented. I'm a strong advocate of separation of concerns and DRY (Don't Repeat Yourself) principles. For instance, in my Algorithm Visualizer project, I structured components clearly to isolate

the visualization logic from the data manipulation logic. I regularly use Git for versioning and review my code for readability and performance before pushing. During my internship at Google, I followed rigorous internal coding standards and review processes, which strengthened my focus on testability, naming conventions, and secure-by-design practices. Writing code that passed both peer and security reviews taught me the value of maintaining consistent, production-quality code from the start.

6. Would you describe yourself as an architect of resilient software? If so, why, and in which sorts of applications?

At my current level, I have been responsible for designing parts of resilient systems, especially during my internship at Google where I created secure VPC environments. While I'm still learning broader architectural patterns, I've made intentional decisions to ensure fault tolerance and modularity in my frontend apps and cloud configurations. I consider resilience a mindset as much as a skill, and I've cultivated that mindset through special attention to edge cases and exception handling.

7. What software products have you yourself led which shipped many releases to multiple customers? What was your role?

As a student developer, I haven't yet led a commercial product with multiple releases to external customers. However, I led the development of the AI Multi-Agent Orchestration tool as a personal project, which involved integrating OpenAI's SDK and managing real-time frontend state. I shared it with my friends and mentors, changed on basis of feedbacks I received, and ensured it was robust enough to simulate collaborative decision-making across agents.

8. What is your most senior role in a software engineering organisation? Describe your span of control, and the diversity of products, functions and teams you led.

My most senior role so far was during my internship at KloudFarm, where I was given end-to-end ownership of a feature-rich Algorithm Visualizer project as a Frontend Developer. I collaborated directly with the lead engineer and was responsible for the architecture, design, development, and Git-based version control.

9. What is your proudest success as an engineering leader?

One of my proudest achievements was during my internship at **Google**, where I contributed to a cloud infrastructure security project involving VPC configuration. I was given the opportunity to work fairly independently on setting up secure access controls and firewall rules. While the scope was guided, it gave me a strong sense of ownership and responsibility. I learned a lot about how large-scale systems are built securely, and it boosted my confidence in working with real-world cloud tools and collaborating with experienced engineers.

10. Outline your thoughts on open source software development. What is important to get right in open source projects? What open source projects have you worked on? Have you been an open source maintainer, on which projects, and what was your role?

Open source is more than just code sharing — it's about building trust and transparency. Documentation, testing, and modular design are crucial to get right in open source projects. I've contributed small tools and components to public GitHub repositories and actively use open source libraries like React, Tailwind CSS, and Redux. While I haven't been a maintainer yet, I aim to contribute more formally, particularly in developer tooling and security.

11. Describe your experience building large systems with many services - web front ends, REST APIs, data stores, event processing and other kinds of integration between components. What are the key things to think about in regard to architecture, maintainability, and reliability in these large

systems?

I haven't yet built a full multi-service system in production, but I've had exposure to system-level design through my internship at **Google**, where I worked on infrastructure and security controls in a cloud environment. From a development perspective, I've primarily focused on **frontend projects**, including an AI orchestration tool where I handled interface logic, data presentation, and user workflows. While I didn't directly implement backend or API layers, working with tools that simulate agent behavior helped me understand the importance of clean data flow, modular design, and predictable interactions between components. I've also spent time learning about system design concepts like separation of concerns, interface contracts, and the value of building maintainable, loosely coupled components that can scale or change independently.

12. How comprehensive would you say your knowledge of a Linux distribution is, from the kernel up? How familiar are you with low-level system architecture, runtimes and Linux distro packaging? How have you gained this knowledge?

My understanding of Linux is foundational but growing. I use Linux for development and administration tasks, understand shell scripting, and use tools like Docker, systemctl, and file permission configurations. I'm currently exploring topics like kernel modules, init systems, and memory management through online courses and reading kernel documentation. I still need to be more fluent with distro packaging systems and system internals.

13. Describe your experience with large-scale IT operations, SAAS, or other running services, in a devops or IS or system administration capacity.

My experience in this area comes from my internship at Google, where I was exposed to cloud security practices at scale. I worked on configuring and securing services using GCP's built-in tools and observed how systems were monitored and audited. I've also simulated smaller-scale setups using Docker and GitHub Actions for CI/CD, which helped me understand deployment workflows and monitoring practices.

14. Describe your experience with public cloud based operations - how well do you understand large-scale public cloud estate management and developer experience?

During my internship on **Google Cloud Platform (GCP)**, I worked on setting up and securing cloud resources. I got hands-on experience with things like **VPCs**, **IAM roles**, and **firewall rules**. I also used tools like **Cloud Armor** and helped with logging and auditing. I learned how important it is to follow best practices like giving only the access that's needed and keeping services isolated for security. I haven't worked on very large cloud systems yet, but I do understand the basics of how cloud infrastructure is organized, and I feel confident working with cloud-native tools in real projects.

15. Outline your thoughts on quality in software development. What practices are most effective to drive improvements in quality?

In my opinion quality in software only comes from a great design, code reviews, testing, and feedback loops. I follow good development habits like using linting tools to catch style or syntax issues early, keeping my code modular, and reviewing it carefully before finalizing. These small steps help avoid bugs and make the code easier for others to read and maintain. In my projects, I've used unit tests (with Jest), functional testing, and logging to catch regressions early. Communication between team members is also key — aligning on what "quality" means helps maintain consistent standards.

16. Outline your thoughts on documentation in large software projects. What practices should teams follow? What are great examples of open source docs?

I think good documentation is really needed for future contributors. Teams should follow practices

like README-first development and inline docstrings. Documentation should be kept close to the code and updated regularly. Open source examples like the Django project and React's docs stand out for their clarity, structure, and thoroughness even though I have no experience with Django but I still aim to follow similar patterns in my own work.

17. Outline your thoughts on user experience, usability and design in software. How do you lead teams to deliver outstanding user experience?

User experience should be considered from the first design discussion. Though I don't have much experience with Backend but even in backend systems, UX manifests as CLI intuitiveness or API consistency. In frontend apps (which is my proficiency) , I focus on clarity, accessibility, and responsiveness. In my projects, I've emphasized mobile responsiveness, clear error states, and simple visual hierarchy. I collaborate closely with designers or use established design systems like Tailwind CSS to maintain a consistent user experience.

18. Outline your thoughts on performance in software engineering. How do you ensure that your product is fast?

I ensure speed through practices like code-splitting in React apps, minimizing API calls, debouncing inputs, and lazy loading components. I also monitor performance using tools like Lighthouse and browser dev tools. Writing efficient algorithms, avoiding unnecessary re-renders, and keeping payloads small are key principles I follow. Performance is a non-negotiable aspect of user satisfaction in my opinion.

19. Outline your thoughts on security in software engineering. How do you lead your engineers to improve their security posture and awareness?

Security must be considered at every stage like from design to deployment. During my GCP internship, I learned the importance of layered security: IAM, VPC Service Controls, and DDoS protection with Cloud Armor. I also stay updated on CVEs and promote awareness of secure coding practices among peers.

20. Outline your thoughts on devops and devsecops. Which practices are effective, and which are overrated?

DevOps is most effective when it empowers developers to own their deployments. I find CI/CD pipelines, infrastructure as code, and automated rollback mechanisms very effective. DevSecOps extends this by integrating security into every pipeline stage, which I support through code scanning and access audits. Overrated practices include over-engineered pipelines or excessive automation without understanding — simplicity with clarity always wins.

Industry Leadership Experience

1. Describe your speaking experience at industry events and conferences.

While I haven't yet had the opportunity to speak at major industry conferences, I have led multiple technical sessions and project presentations in academic settings. During my time as President of the Student Union, I organized and hosted tech awareness sessions and coding workshops for students. I enjoy explaining technical topics and aspire to contribute to developer meetups and open-source communities in the future as a speaker and mentor.

2. Are you a thought leader in any particular area of technology?

At this stage in my career, I wouldn't call myself a thought leader, but I am an active learner and contributor in the areas of frontend engineering and cloud security. I plan to expand my presence in open-source and developer communities to grow into a leadership role in the future.

3. Describe any experience working with startups. What did you draw from that experience that would be relevant for this application?

I interned at KloudFarm, a startup where I developed an interactive Algorithm Visualizer. The startup environment exposed me to fast-paced development, close collaboration with the core team, and iterative problem-solving with limited resources. It taught me to take ownership, move quickly without compromising quality, and wear multiple hats when necessary and if I may not wrong all of them are highly relevant to Canonical's engineering culture.

4. Describe any experience working in a public company. What is important for your colleagues to know about being a public company?

At Google, a public company, I learned the importance of rigorous internal processes, documentation standards, and long-term thinking in software development. Being part of a public company also showed me how decisions are made with broader stakeholder accountability in mind. Engineers need to be aware that their work not only serves immediate users but also represents the company's reliability and reputation in the market.

Education

1. How did you rank in your final year of high school in mathematics? Were you a top student? On what basis would you say that?

Yes, I was among the top students in mathematics during my final year. I consistently scored above 90% in school assessments and was often chosen to represent the school in math competitions. My problem-solving skills and logical reasoning were recognized by both peers and faculty, contributing to my leadership roles within the academic community.

2. How did you rank in your final year of high school, in your home language? Were you a top student? On what basis would you say that?

In my home language, English, I performed very well and was often praised for my communication skills. I scored in the top range during board exams and was frequently selected to deliver speeches and lead student discussions even in my college. These communication skills have helped me significantly in explaining complex ideas clearly in both technical and non-technical settings.

3. Please state your high school graduation results or university entrance results, and explain the grading system used.

I completed my 10th standard under the **Central Board of Secondary Education (CBSE)** with a score of **84.67%**, and my 12th standard under the **Maharashtra State Board** with a score of **82.33%**. In India, both boards use a **percentage-based system out of 100**, and scores above 80% are generally considered very good due to the competitive academic environment. For university admissions, I appeared for the **Kerala Engineering Entrance Examination (KEAM)** and secured a rank of **6724 among several lakh (three hundred thousand+) candidates**. Based on this result, I gained admission to the **Cochin University of Science and Technology (CUSAT)**, which is one of the more reputable and competitive technical universities in India as it is a Government college.

4. Can you make a case that you are in the top 5% in your academic year, or top 1%, or even higher? If so please outline that case.

While my cumulative GPA is around **7.5 out of 10**, which may not place me in the academic top 5%, I strongly believe I stand out in terms of **technical skills, practical experience, and industry readiness**. My selection for internships at highly competitive organizations like **Google** and **KloudFarm** both based on technical merit reflects this. These opportunities typically attract thousands of applicants,

and being chosen indicates that I am in the **top 1% or even 0.1%** in terms of real-world software engineering skills. I've also led key technical projects, actively contributed to development teams, and focused on building practical, production-ready tools which are often not reflected directly in GPA but are critical indicators of engineering potential.

5. What sort of high school student were you? Outside of class, what were your interests and hobbies? What would your high school peers remember you for?

I was a balanced student — strong academically but also deeply involved in extracurricular activities. As the President of the Student Union, I was known for organizing events and voicing student concerns. My peers remember me as approachable, technically skilled, and someone who always volunteered for team initiatives. Outside of class, I enjoyed working on coding puzzles, reading tech blogs, and participating in school-level hackathons. Apart from this I had interest in volleyball as well as chess. There is one interest of mine which is playing games, I have played almost all the games and I still love it.

6. Which university and degree did you choose? What other universities did you consider, and why did you select that one?

I chose to pursue a Bachelor's of Technology in Computer Science at Cochin University of Science and Technology (CUSAT). I selected CUSAT because of its strong engineering faculty, research environment, and affordable, merit-based admission system. The university also has a vibrant student developer community, which appealed to me.

7. Overall, what was your degree result and how did that reflect on your ability? Please help us understand the grading system for your results.

My degree is still in progress, but I've consistently performed well in coursework, labs, and projects. The Indian university grading system typically awards marks out of 10 CGPA or 100 percentage points. I've maintained a strong academic record while balancing internships, leadership roles, and personal projects, which reflects both time management and deep engagement with my field. Even though I am still pursuing my degree I can still work from home and it would not effect my work with Canonical.

8. During all of your education years, from high school to university, can you describe any achievements that were truly exceptional?

One of my standout achievements was being selected for an internship at Google, which is a globally recognized program and highly competitive. Additionally, I led the development of a significant frontend tool at KloudFarm as an intern and served as President of the Student Union during high school — both of which required strong leadership, responsibility, and initiative.

9. What leadership roles did you take on during your education? Did you conceive of, and drive to completion, any initiatives outside of your required classwork?

I served as President of the Student Union in high school, where I led initiatives including student wellness programs, inter-school tech events, and academic tutoring for peers. In university, I initiated peer code review groups and helped organize informal tech meetups. These initiatives were driven by a desire to build community, share knowledge, and improve the academic environment for everyone.

Context

1. Outline your thoughts on the mission of Canonical. What is it about the company's purpose and goals which is most appealing to you? What do you see as risky or unappealing?

Canonical's mission to democratize access to technology through open-source software is inspiring. I admire how Ubuntu made Linux accessible to millions and continues to push boundaries in cloud-native computing, security, and automation. The focus on high-quality, well-maintained open tools is something I deeply align with. One challenge, however, might be the fast pace of open-source innovation — balancing stability with rapid updates can be risky if not handled carefully.

2. Who are Canonical's key competitors, and how should Canonical set about winning?

Canonical's key competitors include Red Hat, SUSE, and cloud-native OS providers like CoreOS and Amazon Linux. Canonical can continue to lead by focusing on developer experience, deep cloud integration, and lightweight infrastructure tools like Multipass and MAAS. Investing more in documentation, community outreach, and seamless multi-cloud workflows will help Canonical remain a strong choice for both enterprises and developers.

3. Why do you most want to work for Canonical?

I want to work at Canonical because of its strong open-source culture, global remote-first environment, and technical rigor. Canonical encourages autonomy, embraces innovation, and operates at the intersection of systems engineering, devops, and security — all areas I'm passionate about. I see it as a place where I can contribute meaningfully, grow technically, and work alongside a mission-driven team and there is one more reason which is Canonical can still hire me as a currently pursuing degree student.

4. What would you most want to change about Canonical?

One area of improvement could be making Canonical more visible in the mainstream developer ecosystem. While Ubuntu is widely used, Canonical's engineering tools and contributions aren't always spotlighted in broader developer communities. I'd love to see more developer showcases, beginner-friendly initiatives, and outreach events to build stronger community engagement.

5. What gets you most excited about this role?

What excites me most is the opportunity to write Python code that impacts global infrastructure while collaborating across teams and time zones. I'm particularly excited by the idea of working on tools that are used and appreciated by the global developer community.